

PVsyst - Simulation report

Grid-Connected System

Project: C7 bilhari

Variant: New simulation variant

No 3D scene defined, no shadings

System power: 10.50 kWp

Bilahri C7 - India



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PVsyst V8.1.3

VCO, Simulation date:
17/07/26 12:16
with V8.1.3

Project summary

Geographical Site

Bilahri C7

India

Situation

Latitude 23.1261 °(N)
Longitude 79.9662 °(E)
Altitude 396 m
Time zone UTC+5.5

Project settings

Albedo 0.20

Weather data

Bilahri C7

Meteonorm 9.0 dll, Sat=100% - Synthetic

System summary

Grid-Connected System

Orientation #1

Fixed plane

Tilt/Azimuth 29.3 / 0 °

System information

PV Array

Nb. of modules

Pnom total

No 3D scene defined, no shadings

Near Shadings

no Shadings

28 units

10.50 kWp

Inverters

Nb. of units

Total power

Pnom ratio

2 units

10 kWac

1.05

User's needs

Unlimited load (grid)

Results summary

Simulation step Hourly

Produced Energy 16662 kWh/year Specific production 1587 kWh/kWp/year Perf. Ratio PR 80.23 %

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General parameters

Grid-Connected System

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Simulation step

Hourly

Orientation #1

Fixed plane

Tilt/Azimuth 29.3 / 0 °

Models used

Transposition Perez
Diffuse Perez, Meteonorm
Circumsolar separate

Horizon

Free Horizon

Near Shadings

no Shadings

User's needs

Unlimited load (grid)

PV Array Characteristics

PV module

Manufacturer Generic
Model LG 375 N1C-A6
(Original PVsyst database)
Unit Nom. Power 375 Wp
Number of PV modules 28 units
Nominal (STC) 10.50 kWp
Modules 4 units x 7 In series

At operating cond. (50°C)

Pmpp 9.63 kWp
U mpp 223 V
I mpp 43 A

Total PV power

Nominal (STC) 11 kWp
Total 28 modules
Module area 50.8 m²
Cell area 46.3 m²

Inverter

Manufacturer Generic
Model Sunny Tripower 5.0-3SE-40
(Original PVsyst database)
Unit Nom. Power 5.00 kWac
Number of inverters 4 * MPPT 50% 2 units
Total power 10.0 kWac
Operating voltage 150-800 V
Pnom ratio (DC:AC) 1.05
No power sharing between MPPTs

Total inverter power

Total power 10 kWac
Number of inverters 2 units
Pnom ratio 1.05

Array losses

Thermal Loss factor

Module temperature according to irradiance
Uc (const) 20.0 W/m²K
Uv (wind) 0.0 W/m²K/m/s

DC wiring losses

Global array res. 84 mΩ
Loss Fraction 1.50 % at STC

Module Quality Loss

Loss Fraction -0.75 %

Module mismatch losses

Loss Fraction 2.00 % at MPP

IAM loss factor

Incidence effect (IAM): Fresnel, AR coating, n(glass)=1.526, n(AR)=1.290

0°	30°	50°	60°	70°	75°	80°	85°	90°
1.000	0.999	0.987	0.963	0.892	0.814	0.679	0.438	0.000



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Main results

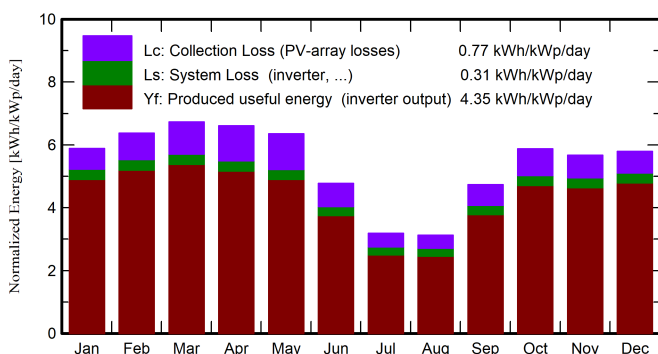
Simulation step

Hourly

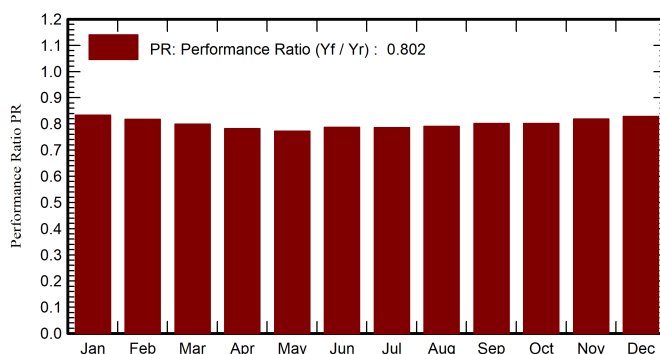
System Production

Produced Energy 16662 kWh/year Specific production 1587 kWh/kWp/year
Perf. Ratio PR 80.23 %

Normalized productions (per installed kWp)



Performance Ratio PR



Balances and main results

	GlobHor	DiffHor	T_Amb	GlobInc	GlobEff	EArray	E_Grid	PR
	kWh/m ²	kWh/m ²	°C	kWh/m ²	kWh/m ²	kWh	kWh	ratio
January	133.3	44.63	17.00	182.6	179.9	1702	1598	0.833
February	142.3	45.22	20.80	178.5	175.7	1628	1531	0.817
March	188.2	65.79	26.00	208.7	204.9	1861	1752	0.800
April	201.4	74.66	31.20	198.3	193.9	1731	1628	0.782
May	218.8	84.01	34.70	197.1	192.2	1701	1597	0.772
June	163.7	93.78	31.70	143.2	139.2	1275	1184	0.787
July	111.4	71.81	27.50	98.9	96.0	899	817	0.786
August	104.0	70.91	26.60	96.8	94.1	886	803	0.790
September	137.6	75.33	27.30	142.0	138.9	1285	1194	0.801
October	155.6	59.39	25.90	182.2	179.0	1636	1533	0.802
November	130.7	52.90	21.80	170.1	167.7	1562	1463	0.819
December	127.3	40.98	18.20	179.6	177.1	1665	1562	0.828
Year	1814.0	779.41	25.74	1977.9	1938.5	17832	16662	0.802

Legends

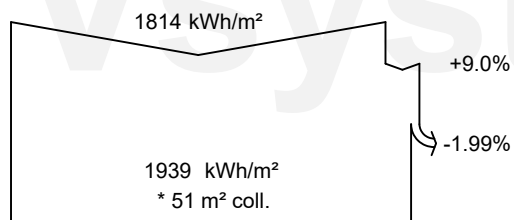
GlobHor	Global horizontal irradiation	EArray	Effective energy at the output of the array
DiffHor	Horizontal diffuse irradiation	E_Grid	Energy injected into grid
T_Amb	Ambient Temperature	PR	Performance Ratio
GlobInc	Global incident in coll. plane		
GlobEff	Effective Global, corr. for IAM and shadings		



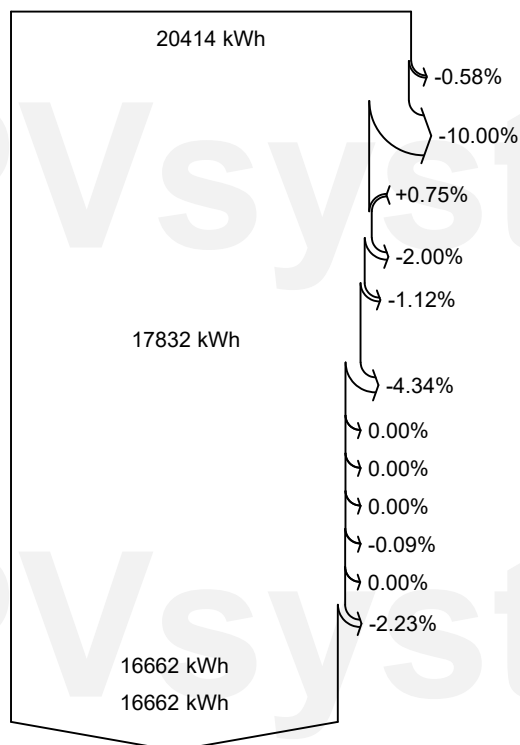
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Loss diagram



efficiency at STC = 20.74%



Global horizontal irradiation
Global incident in coll. plane

IAM factor on global

Effective irradiation on collectors

PV conversion

Array nominal energy (at STC effic.)

PV loss due to irradiance level

PV loss due to temperature

Module quality loss

Module array mismatch loss

Ohmic wiring loss

Array virtual energy at MPP

Inverter Loss during operation (efficiency)

Inverter Loss over nominal inv. power

Inverter Loss due to max. input current

Inverter Loss over nominal inv. voltage

Inverter Loss due to power threshold

Inverter Loss due to voltage threshold

Night consumption

Available Energy at Inverter Output

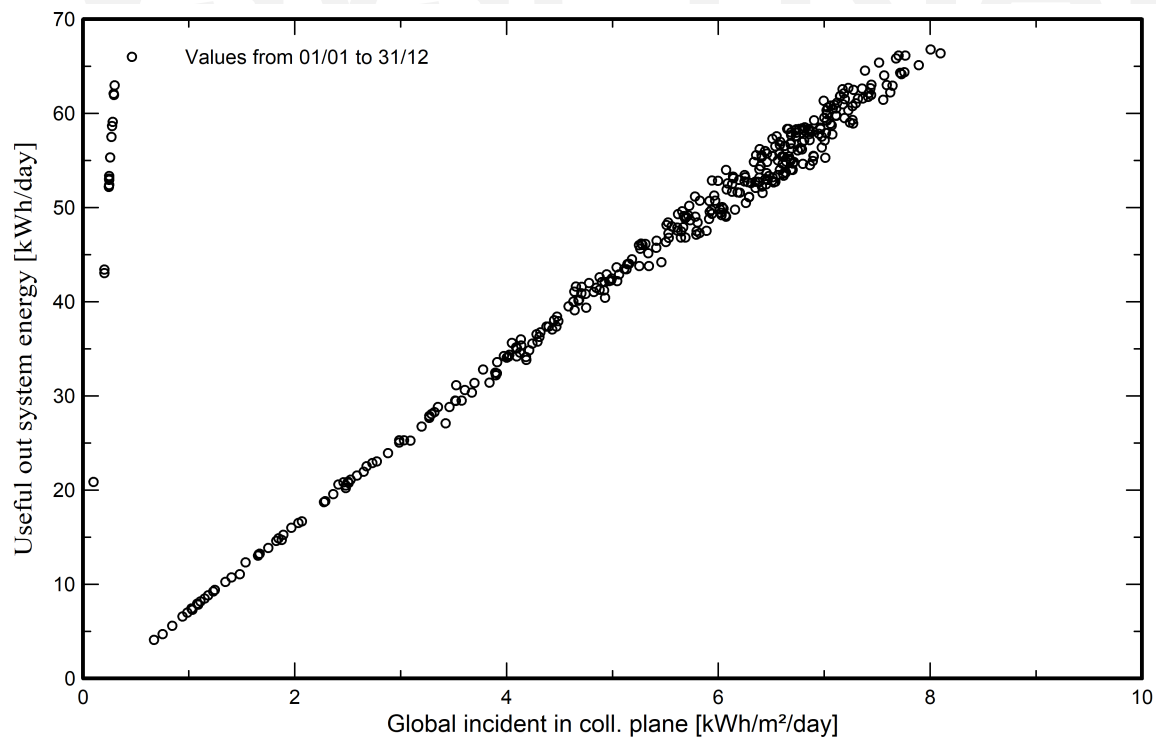
Energy injected into grid



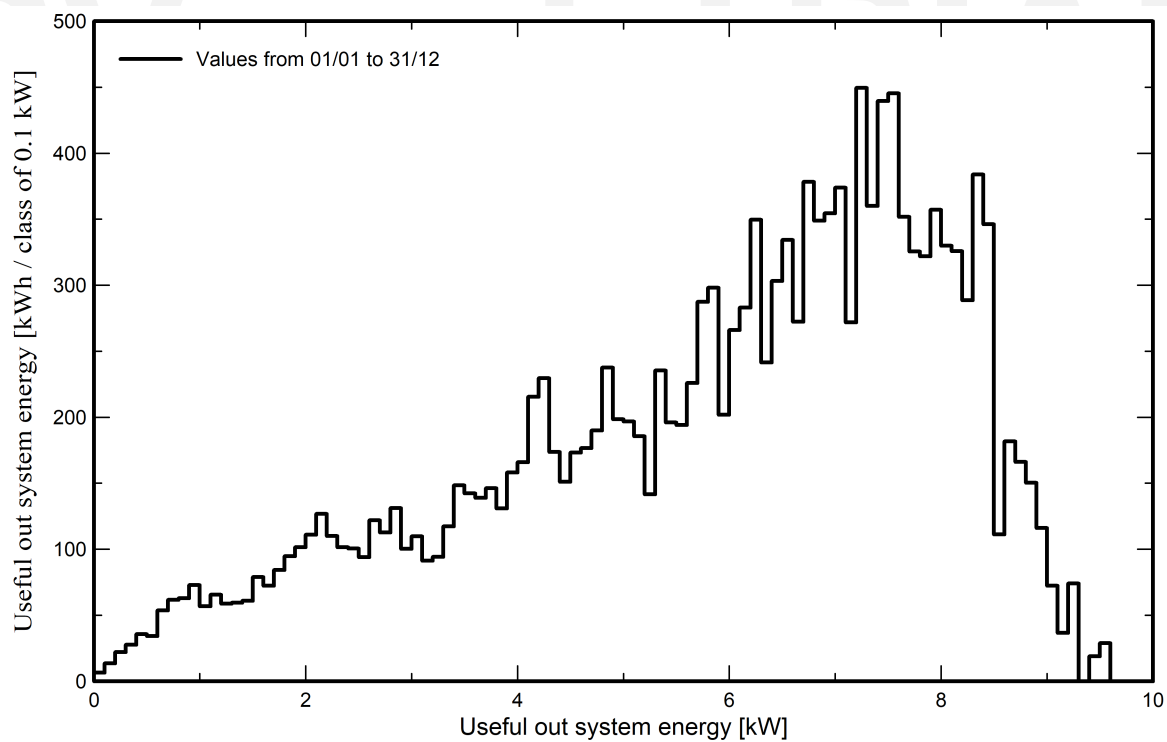
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Predef. graphs
Daily Input/Output diagram



System Output Power Distribution

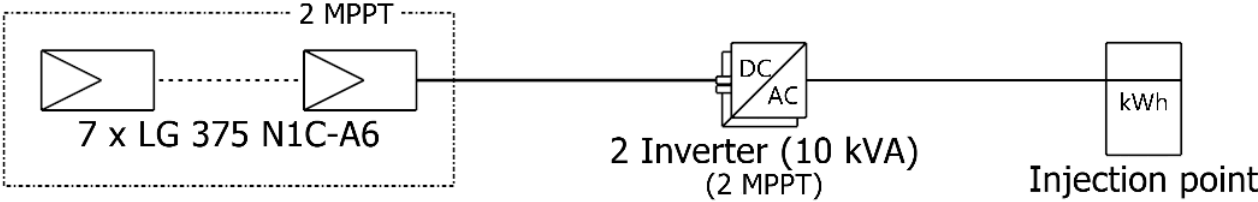




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Single-line diagram



PV module	LG 375 N1C-A6
Inverter	Sunny Tripower 5.0-3SE-40
String	7 x LG 375 N1C-A6

C7 bilhari

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